Assignment - 1

1) write info about diff. types of routing protocol

Ans → A routing Protocol is a set of rules used by routers to determine the best Path for forwarding data packets from Source to destination

(i) RIP (Routing information protocol)
- uses hop count as its metric
- It is a distance vector routing protocol
- maximum 15 hops are allowed 16 means unreachable
- Simple but slower for large networks

(ii) OSPF (Open Shortest Path first)
- It is link state routing Protocol
- uses Dijkstra's Shortest path algorithm
- uses cost as its metric
- fast convergence and Suitable for large networks

(iii) BGP (Border gateway Protocol)
- It is a path vector routing Protocol
- used to exchange routing information between difft Autonomous System (AS)

- It is the primary routing protocol of the internet

(iv) IS - IS (Intermediate system to Intermediate system)
- It is a link state routing protocol
- uses the SPF Algorithm
- Supports hierarchical routing

2) Compare OSPF, IGRP, EIGRP, RIP, VLAN

features	OSPF	IGRP	EIGRP	RIP	VLAN
full form	Open Shortest path first	Interior gateway Routing Protocol	Enhance Interior gateway Routing Protocol	Routing Information protocol	virtual Local Area network
Type	Link state	distance vector	Advance distance vector	distance vector	network segmentation
metric	COST	composite metric	composite metric	Hop count	VLAN ID
Algorithm	SPF	Bellman Ford	DUAL	Bellman ford	No algo
maximum hop count	No fixed 15 hop Limit	255	100	15	Not applicable

Convergence	fast	Slow	very fast	Slow	Not applicable
Standard/use	open	Cisco Proprietary Obsolete	Cisco original	open	IEEE 802.1Q
uses	Medium/Large Networks	older Cisco Networks	enterprise Networks	Small networks	Seperate broadcast domain

3) Different Types of Networking devices

Answer

(i) Hub → connects multiple devices in a network
→ Broadcast incoming data to all connected devices
→ works mainly at Layer 1
→ Less secure and efficient than switch

(ii) Switch → Connects multiple device with in LAN
→ Uses MAC addresses to forward frames
→ works mainly at Layer 2
→ More efficient than a hub

(iii) Router → connects different networks
→ Uses IP addresses to determine

the best Path
→ works at Layer 3

(iv) Modem → converts digital signals to signal suitable for transmission over the communication medium and back again
→ commonly used to connect a network to an ISP

(v) Repeater → Recieves and regenerates weak signals
→ extends the communication distance of a network
→ works at Layer 1

(vi) Bridge → connects two LAN segments
→ uses MAC address to filter traffic
→ works at Layer 2

4) write info about storage devices and file system

Answer

A. Storage devices → hardware devices used to Store data, program, and OS Permanent or temporary

(i) HDD - Hard disk drive

- uses magnetic disks to store data
- usually provides large storage capacity
- Cheaper than SSD

(ii) SSD - Solid State drive

- uses flash memory
- has no moving parts
- Much faster than HDD

(iii) USB Flash drive

- Portable Storage device
- uses flash memory
- Connects through USB

(iv) Memory card

- Small flash Storage device
- commonly used in Phone, cameras, and other portable devices
- example : SD, micro SD

B. File System → It is a method used by an OS to organize, store, name and retrieve file on a storage device

(i) FAT32
- Older and widely compatible file system
- Maximum individual file size is approximately 4 GB
- Used in USB drives and memory cards

(ii) exFAT
- designed for flash storage and removable devices
- Supports files larger than 4 GB
- Compatible with many OS

(iii) NTFS
- Common file system used by windows
- Supports permissions, encryptions, Journaling, and Large files

(iv) ext4

- common Linux file system
- Supports Journaling & large files
- widely used for Linux operating system

(v) APFS

- Modern file system developed by Apple
- Optimized for SSDs and flash storage
- used in modern macOS and Apple devices